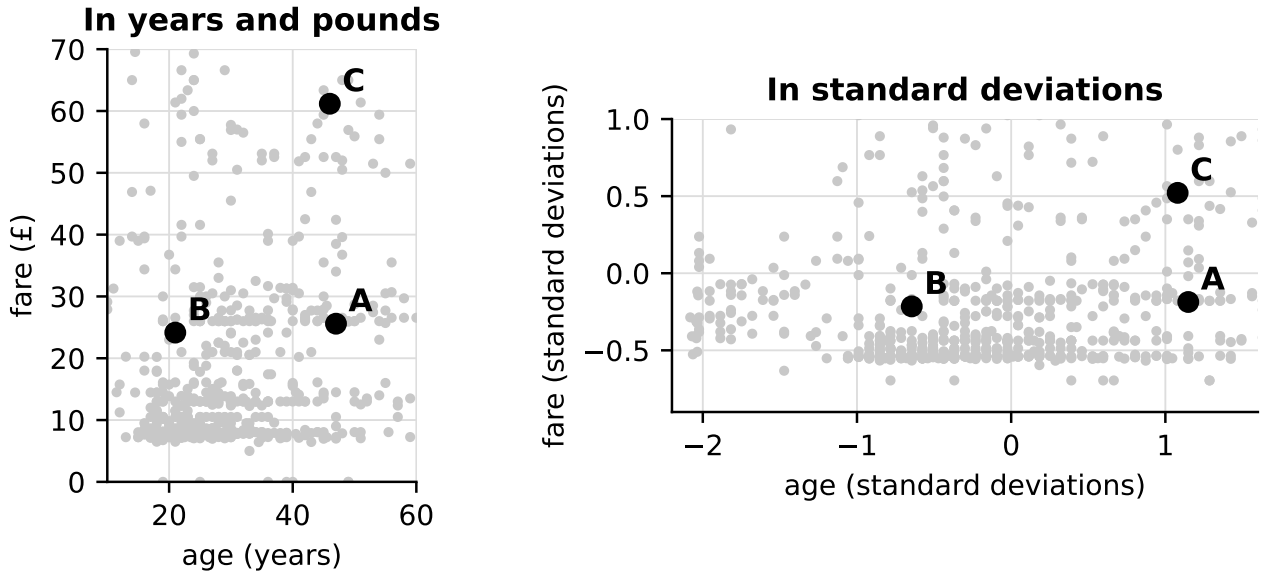
DATA 202 — Scaling, Encoding and Pipelines · Week 9 Monday

Same passengers as today's slides: the 1,043 *Titanic* passengers with age, fare and port recorded. Your professor will tell you when to do each part.

1 · Who is closest?



passenger	class	age	fare
A — Colley, Mr. Edward	1st	47	£25.59
B — Davies, Mr. John	3rd	21	£24.15
C — Chaffee, Mr. Herbert	1st	46	£61.18

1. On **both** plots, draw the lines A–B and A–C, and circle the passenger closer to A.
2. **Talk with your neighbor:** the two plots disagree. Using the table too, which answer would you trust to find a passenger like A, and why?

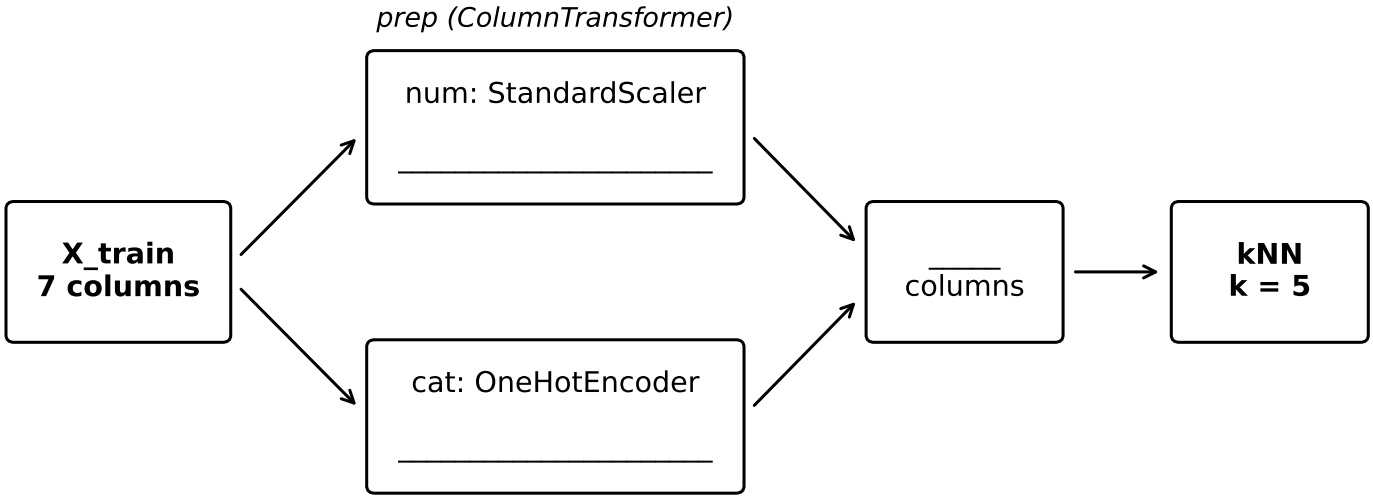
2 · One-hot encode four passengers

One-hot: one 0/1 column per category, with a 1 only in the row's own category.

#	sex	embarked	sex_male	Cherbourg	Queenstown	Southampton
1	male	Southampton	_____	_____	_____	_____
2	female	Cherbourg	_____	_____	_____	_____
3	male	Queenstown	_____	_____	_____	_____
4	female	Southampton	_____	_____	_____	_____

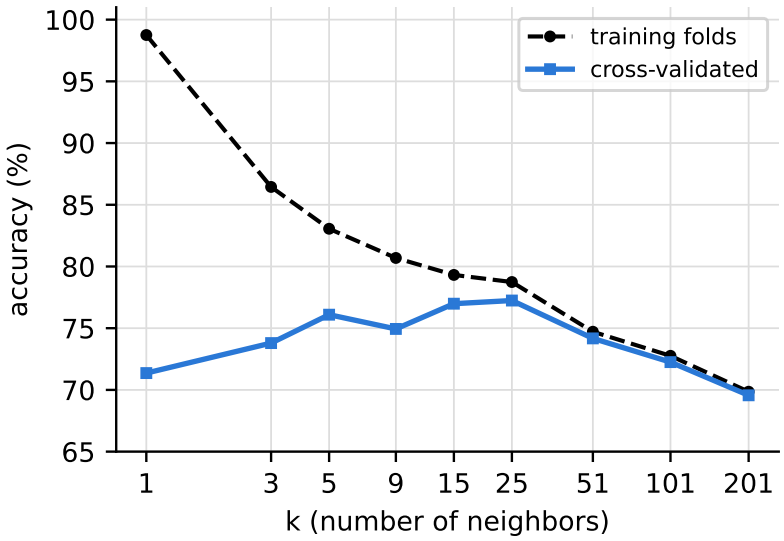
Talk with your neighbor: one-hot makes every port *equally far* from every other. Is that fair to Southampton, Cherbourg and Queenstown? What would you want to know about them?

3 · Follow the pipe



In each branch, write the columns that go in. In the right box, write how many columns come out into kNN.

4 · Choose k



Accuracy of the whole pipeline for each k, averaged over 5 folds of the training passengers: on the folds it trained on, and on the fold it was scored on (cross-validated). The test passengers were not used.

1. **Circle** the best k.
2. **Label** the overfitting side and the underfitting side.